8.8.1 Equations of Motion for the NHL Piston

In the framework of [199], rewriting the equations of motion for the physical variables q and p, the dynamics takes the form

$$\begin{aligned}\dot{\boldsymbol{q}} &= \boldsymbol{M}^{-1}\boldsymbol{p} + \eta\boldsymbol{q},\\ \dot{\boldsymbol{p}} &= -\nabla U(\boldsymbol{q}) - \eta\boldsymbol{p} - \xi\boldsymbol{p},\\ \dot{V} &= \eta V,\end{aligned}$$

with auxiliary variables η and ξ governed by additional dynamical equations of motion

$$\begin{aligned}\dot{\xi} &= h_{\mathrm{T}} \stackrel{\mathrm{def}}{=} (\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T})/\mu_{\mathrm{T}},\\ \dot{\eta} &= h_{\mathrm{P}} \stackrel{\mathrm{def}}{=} (V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \mathrm{P}V + \mathrm{k_B T})/\mu_{\mathrm{P}},\end{aligned}$$

where

$$\Pi(\boldsymbol{q},\boldsymbol{p},V) \stackrel{\mathrm{def}}{=} \frac{1}{V}\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \boldsymbol{q}\cdot\nabla U(\boldsymbol{q}) - V\frac{\partial}{\partial V}U(\boldsymbol{q})\right),$$

is the expression for the pressure.

Now let us examine the distribution conserved by the dynamics. First, we observe

$$\rho_1 = \varphi_1(\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})) \times \psi(V,\xi,\eta)$$

is stationary for the Liouvillian operator for the vector field $(\boldsymbol{p}, -\nabla U, 0, 0, 0)$.

The function ψ can be chosen as any smooth function that integrates out with respect to V, ξ, η over $[0,\infty] \times \mathbb{R} \times \mathbb{R}$. We already know from our discussion of Nosé-Hoover that $\dot{\boldsymbol{p}} = -\xi\boldsymbol{p}$; $\dot{\xi} = (\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} - \mathrm{N_d k_B T})/\mu_{\mathrm{T}}$ will have as stationary states functions of the form

$$\rho_2 = \exp(-\beta[\boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p} + \mu_{\mathrm{T}}\xi^2/2])\psi_2(\boldsymbol{q},\eta,V).$$

Finally the remaining terms $\dot{\boldsymbol{q}} = \eta\boldsymbol{q}, \dot{\boldsymbol{p}} = -\eta\boldsymbol{p}$, $\dot{V} = \eta V$, $\dot{\eta} = (V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \mathrm{P}V + \mathrm{k_B T})/\mu_{\mathrm{P}}$ give rise to the Liouvillian

$$\mathcal{L}^\dagger\rho = -\eta\nabla_{\boldsymbol{q}}\cdot(\boldsymbol{q}\rho) + \eta\nabla_{\boldsymbol{p}}\cdot(\boldsymbol{p}\rho) - \eta\frac{\partial}{\partial V}(V\rho) - \frac{\partial}{\partial\eta}[\rho(V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \mathrm{P}V + \mathrm{k_B T})/\mu_{\mathrm{P}}].$$

If ρ has the form $\rho = \exp(-\beta\mu_{\mathrm{P}}\eta^2/2) \times \psi(\boldsymbol{q},\boldsymbol{p},V,\xi)$ then eliminate η, reducing $\mathcal{L}^\dagger\rho = 0$ to

$$-\nabla_{\boldsymbol{q}} \cdot (\boldsymbol{q}\psi) + \nabla_{\boldsymbol{p}} \cdot (\boldsymbol{p}\psi) - \frac{\partial}{\partial V}(V\psi) + \beta(V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \mathrm{P}V + \mathrm{k_B T})\psi = 0.$$

This results in

$$-\boldsymbol{q} \cdot \nabla_{\boldsymbol{q}}\psi + \boldsymbol{p} \cdot \nabla_{\boldsymbol{p}}\psi - \psi - V\frac{\partial\psi}{\partial V} + (\beta V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \beta\mathrm{P}V + 1)\psi = 0. \tag{8.74}$$

A solution (the desired one) is given by

$$\psi \propto \exp(-\beta(H + \mathrm{P}V)),$$

where substituting into (8.74) gives

$$\Big[\beta\boldsymbol{q} \cdot \nabla U(\boldsymbol{q}) - \beta\boldsymbol{p} \cdot \boldsymbol{M}^{-1}\boldsymbol{p} - 1 + \beta V\frac{\partial U}{\partial V} + \beta\mathrm{P}V + (\beta V\Pi(\boldsymbol{q},\boldsymbol{p},V) - \beta\mathrm{P}V + 1)\Big]\psi = 0,$$

which is quickly checked to be true, and hence the dynamics preserves $\rho = \exp(-\beta\mu_{\mathrm{P}}\eta^2/2) \times \psi$.

It is difficult to see how their could be other solutions ψ, but we know from the Nosé-Hoover case that this must be possible. We improve the robustness of the method by introducing Ornstein-Uhlenbeck type stochastic processes in the two auxiliary variables ξ and η, yielding the system

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p} + \eta\boldsymbol{q}, \tag{8.75}$$

$$\dot{\boldsymbol{p}} = \boldsymbol{F}(\boldsymbol{q}) - \eta\boldsymbol{p} - \xi\boldsymbol{p}, \tag{8.76}$$

$$\dot{V} = \eta V, \tag{8.77}$$

$$\mathrm{d}\xi = [h_{\mathrm{T}} - \gamma_{\mathrm{T}}\xi]\,\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma_{\mathrm{T}}/\mu_{\mathrm{T}}}\mathrm{dW_T}, \tag{8.78}$$

$$\mathrm{d}\eta = [h_{\mathrm{P}} - \gamma_{\mathrm{P}}\eta]\,\mathrm{d}t + \sqrt{2\mathrm{k_B T}\gamma_{\mathrm{P}}/\mu_{\mathrm{P}}}\mathrm{dW_P}, \tag{8.79}$$

where $\gamma_{\mathrm{T}},\ \gamma_{\mathrm{P}} \geq 0$ are two parameters governing the strength of the noise.

8.8.2 *Discretization of the NHL Piston*

Observations in Chap. 7 and our experience with the Nosé-Hoover-Langevin method lead us to propose splitting methods to separate the Ornstein-Uhlenbeck processes